TechWhiz@DravidianLangTech 2024: Fake News Detection Using Deep Learning Models

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01 Introduction

- Detection methods scrutinize social media text meticulously, aiming to distinguish between original and misleading content, combating the spread of fake news.
- The task focuses on developing models to classify fake news, particularly in Malayalam-language articles, aiming to address the challenge of misleading information dissemination.
- Enhancing NLP promotes informed, reliable digital space, upholds integrity, and encourages responsible communication.

02 Objective

- To develop robust systems for detecting fake news
- To develop a Fake News detection model
- To utilize transformer models

03 Dataset Description

- Task 1: Binary classification (fake or original) with provided datasets for training, evaluation, and test.
- Task 2: Categorization into five fake news categories (Half True, Mostly False, Mostly True, Partly False, and False) using training and test datasets.
- Task 1 had 3,257 training instances, while Task 2 had 1,669 instances, indicating data distribution imbalances.

Category	Training Dataset	Evaluation Dataset
Fake	1,599	1,658
Original	406	409

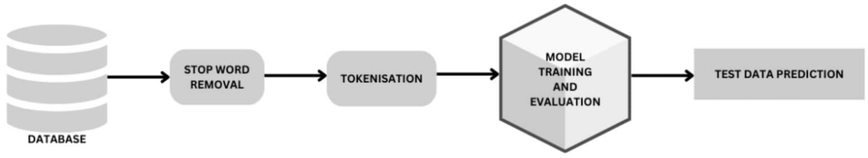
Table 1: Data Distribution for task 1

Category	Training Dataset
Half true	141
Mosty false	239
Mostly true	1
Partly false	42
False	1,246

Table 2: Data Distribution for task 2

04 Methodology

- The labels were converted into integer categorical values to facilitate input to transformer models.
- The methodology involved preprocessing data by removing unwanted characters and training models on datasets from the task organizers.
- Task 1 categorizes social media text to distinguish original from fake news. Task 2 targets Malayalam articles, classifying misinformation types.
- Task 1 utilizes mBERT, ALBERT, and XMLRoBERTa; Task 2 employs XMLRoBERTa, mBERT, and BERT.
- For training, 10 epoch was used formBERT, ALBERT, XMLRoBERTa, and BERT.
- The project architecture is given below:



05 Inference

- Misclassification in fake news detection can occur due to data imbalances, language complexities, ambiguity, lack of context, and biases in data or model.
- The metric values achieved for Task 1 and Task 2 by the three different models were obtained when they were evaluated on the test dataset provided for the shared task..
- The F1 score of the other models in the task indicates an issue related to imbalance, showing higher rates of false positives and negatives, thus emphasizing the significance of data augmentation.

Model	F1-Score	Accuracy
XLMRoBERTa	0.84	0.84
ALBERT	0.56	0.58
mBERT	0.84	0.84

Table 3: Performance Score for Task 1

Model	F1-Score	Accuracy
XLMRoberta	0.21	0.74
BERT	0.11	0.25
mBERT	0.16	0.63

Table 4: Performance Score for Task 2

06 Conclusion

- Outperforming mBERT and ALBERT in Task 1, XMLRoBERTa demonstrates its superior ability in detecting fake news in Dravidian languages.
- Across Task 2, XMLRoBERTa consistently outshines mBERT and BERT, particularly in classifying Malayalam articles, showcasing its effectiveness in handling linguistic complexities.
- XMLRoBERTa combats fake news, advancing Dravidian language tech. Emphasizes reliability, innovation in NLP, fostering responsible digital communication.

Reference

1. Maryam Heidari, Samira Zad, Parisa Hajibabaee, Masoud Malekzadeh, SeyyedPooya HekmatiAthar, Ozlem Uzuner, and James H Jones. 2021. BERT Model for Fake News Detection Based on Social Bot Activities in the COVID-19 Pandemic. In 2021 IEEE 12th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), pages 0103–0109. IEEE.

2. Mahibha Jerin, Kayalvizhi Sampath, and Thenmozhi Durairaj. 2021. Sentiment Analysis using Cross Lingual Word Embedding Model. In Proceedings of Forum for Information Retrieval Evaluation-FIRE 2021.

3. Malliga Subramanian, Bharathi Raja Chakravarthi, Kogilavani Shanmugavadivel, Santhiya Pandiyan, Prasanna Kumar Kumaresan, Balasubramanian Palani, Premjith B, Sandhiya Raja, Vanaja, Mithunajha S, Devika K, Hariprasath S.B, Haripriya B, and Vigneshwar E. 2024. Overview of the Second Shared Task on Fake News Detection in Dravidian Languages. In Proceedings of the Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages, Malta. European Chapter of the Association for Computational Linguistics.

4.Chenguang Song, Nianwen Ning, Yunlei Zhang, and Bin Wu. 2021. Knowledge Augmented Transformer for Adversarial Multidomain Multiclassification Multimodal Fake News Detection. Neurocomputing, 462:88–100.

5. Xingang Wang, Xiaomin Li, Xiaoyu Liu, and Honglu Cheng. 2022. Using ALBERT and Multi-modal Circulant Fusion for Fake News Detection. In 2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pages 2936–2942. IEEE.